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EXTENDED ABSTRACT

ClimaGuard: A Transparent AI Framework for Assessing Solar Radiation Management

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1. Introduction

As global warming accelerates, the urgency to explore all potential pathways for mitigating catastrophic climate change renewed interest in Solar Radiation Management (SRM), including stratospheric aerosol injection (SAI) and marine cloud brightening (MCB). These techniques offer theoretical means to rapidly modulate global temperature [4, 3, 5]. However, the complex, non-linear interactions governing atmospheric processes introduce profound uncertainties regarding regional climate disruptions, environmental side effects (e.g., ozone depletion and altered precipitation patterns), and unintended feedback.

Current evaluation of SRM interventions relies on computationally expensive numerical climate models [2]. These computational constraints limit ensemble size, spatial resolution, and forecast horizon, restricting robust uncertainty quantification, particularly for extreme events, and the development of adaptive, regionally specific governance frameworks.

To bridge the gap between fidelity and computational feasibility, we propose *ClimaGuard*, an open-source framework that integrates deep learning (DL) and explainable artificial intelligence (XAI). ClimaGuard aims to deliver accelerated, high-resolution climate predictions while preserving physical consistency and enabling transparent, interpretable analysis essential for trustworthy climate intervention research.

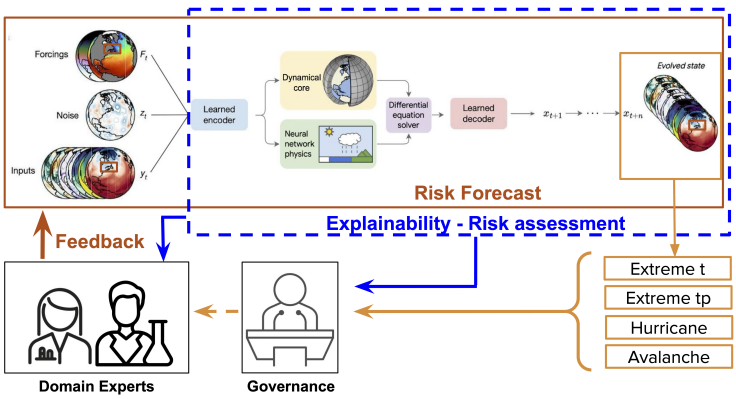

Figure 1. *Schematic overview of the ClimaGuard framework..*

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2. Methodology

ClimaGuard leverages NeuralGCM [1], a hybrid atmospheric model that combines a numerical core with machine learning components. As illustrated in Figure 1, the probabilistic model ingests high-dimensional climate state variables (y_t), stochastic noise components (z_t), and explicit forcing parameters (F_t). The inputs are processed through an encoder and integrated by a combination of a differentiable dynamical core and a neural physics module for parametrization.

Our training data is derived from CMIP6 ARISE-SAI configurations and data from recent runs of the Geoengineering Model Intercomparison Project (GeoMIP), simulating a variation of MCB and SAI scenarios [2, 6]. For validation, we assess predictive skill during large volcanic eruptions, serving as natural analogs for stratospheric aerosol forcing.

Daily model output and 6-hourly ERA5 reanalysis data are processed at a 1° spatial resolution and regridded to 0.7°. Forecasts are produced using 60-day-rollout simulations with periodic SST updates derived from GeoMIP outputs.

NeuralGCM is adapted for SRM via a multi-level strategy: First, we fine-tune the ML components using SRM-specific simulations and associated aerosol and radiative forcing fields. If predictive skill falls below high-resolution benchmarks (e.g., HRES, ERA5), we secondly extend the physical core by incorporating simplified aerosol transport or enhanced stratospheric circulation terms. We evaluate the 60-day forecast horizon using standard metrics (RMSE, ACC) and physical consistency diagnostics.

3. Risk Assessment, Transparency, and Future Impact

The risk assessment targets the frequency and intensity of extremes, including temperature, precipitation, and tropical cyclones, and enables counterfactual analyses, such as the regional climate impacts of shifting aerosol injection altitudes by 1 km. To address the "black-box" nature of these models, ClimaGuard integrates XAI, ensuring physically meaningful predictions. This includes mechanistic interpretability to verify adherence to physical laws and attribution mapping to identify drivers of regional uncertainty and risk.

Ultimately, ClimaGuard aims to overcome the computational bottleneck in SRM research by combining the speed and scalability of AI with the rigor and transparency of physically informed modeling and XAI techniques. Thus, this toolbox provides climate scientists, engineers, and policymakers with an agile, high-fidelity tool for exploring the localized and global impacts of SRM strategies.

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